

Movie Recommendation System Using Review Texts

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Abstract—Traditional movie recommendation systems predominantly rely on collaborative filtering or metadata tags, often overlooking the rich feedback embedded in written user reviews. This project presents a semantic recommendation engine designed to interpret natural language queries using review texts. Utilizing a merged dataset of over 1.4 million Rotten Tomatoes reviews and movie metadata, we implement a progressive pipeline ranging from lexical TF-IDF baselines to transformer-based retrieval architectures. The final system employs a hybrid approach combining Sentence-BERT for efficient candidate retrieval and a Cross-Encoder for high-precision re-ranking. Experimental results demonstrate that the proposed two-stage architecture significantly improves ranking quality over baseline approaches.

I. INTRODUCTION

Traditional recommendation systems often rely heavily on numerical ratings or collaborative filtering. While effective, these approaches fail to capture nuanced opinions, emotional tone, and contextual feedback present in written reviews.

This project aims to design a text-driven movie recommendation system that leverages Natural Language Processing (NLP) techniques to analyze review content directly. By incorporating semantic retrieval models and transformer-based re-ranking, the system enables expressive user queries and improved recommendation relevance.

II. RELATED WORK AND BACKGROUND

Content-based and collaborative filtering methods dominate traditional recommender systems, but suffer from cold-start issues and limited semantic understanding. Recent advances in transformer architectures, such as BERT and Sentence-BERT, have enabled semantic search systems that retrieve documents based on meaning rather than keyword overlap.

Our work builds on these developments by applying a hybrid bi-encoder and cross-encoder architecture to movie recommendation using review text as the primary signal.

III. METHODOLOGY AND SYSTEM ARCHITECTURE

A. Dataset Construction and Preprocessing

The system was built by merging two complementary datasets:

- 1) Rotten Tomatoes Movie Reviews (Kaggle): Provided over 1.4 million reviews with sentiment labels and reviewer metadata.

- 2) Movies Dataset (Hugging Face): Provided essential metadata such as genres, cast, and popularity scores.

Preprocessing Pipeline:

- 1) Merging: Datasets were merged on movie titles and IDs to create a unified corpus.
- 2) Text Cleaning: Review texts underwent standard cleaning, including HTML tag removal and lowercasing. The WordNet lemmatizer was applied to standardize textual features.
- 3) Score Normalization: Original scores were mapped to a uniform 0-10 scale to standardize ratings across different grading systems.
- 4) Feature Aggregation: A combined_text feature was engineered for each movie, concatenating the Title, Overview, and all associated Review Texts into a single rich semantic block.

B. Algorithm Evolution

The system was developed incrementally:

- 1) **Popularity Baseline:** Genre filtering with ranking by average score.
- 2) **TF-IDF Retrieval:** Lexical similarity using TF-IDF (30k features, 1–2 n-grams).
- 3) **Bi-Encoder Retrieval:** Sentence-BERT embeddings (all-MiniLM-L6-v2) for dense semantic retrieval.
- 4) **Two-Stage Hybrid:** Bi-Encoder retrieval followed by Cross-Encoder re-ranking (ms-marco-MiniLM-L-6-v2) with heuristic genre and score filters.

C. Explainability via LLM

To enhance interpretability, Google Gemini 2.0 Flash was used to generate natural-language explanations based on retrieved movies and review summaries. The LLM does not influence ranking and is used solely for explanation generation.

IV. EXPERIMENTS AND RESULTS

A. Exploratory Data Analysis

Genre distribution analysis revealed dominance of Drama and Comedy, motivating strict genre filtering. Score distributions peaked between 6 and 8, motivating heuristic thresholds to distinguish highly-rated and poorly-rated movies.

B. Comparative Evaluation of Models

Figure 1 and Table I summarize the comparative performance of the three retrieval models evaluated at $K = 10$.

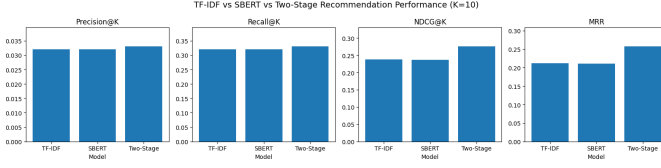


Fig. 1. Comparison of TF-IDF, SBERT, and Two-Stage Recommendation Models at $K = 10$ across Precision@K, Recall@K, NDCG@K, and MRR.

TABLE I
QUANTITATIVE RETRIEVAL PERFORMANCE COMPARISON AT $K = 10$

Model	Precision@K	Recall@K	NDCG@K	MRR
TF-IDF	0.032	0.32	0.238	0.212
SBERT	0.032	0.32	0.236	0.211
Two-Stage	0.033	0.33	0.275	0.257

The two-stage system outperformed both baselines in ranking-sensitive metrics such as NDCG and MRR, demonstrating improved relevance ordering.

V. DISCUSSION

The results highlight the trade-off between efficiency and precision. While bi-encoder retrieval is fast, cross-encoder re-ranking substantially improves relevance. Review text provides richer semantic signals than metadata alone, enabling expressive queries such as nuanced sentiment or thematic preferences.

VI. CONCLUSION AND FUTURE WORK

We presented a semantic movie recommendation system leveraging review text and transformer-based retrieval models. The hybrid two-stage architecture significantly improves ranking quality over TF-IDF and bi-encoder baselines.

Future work includes personalization using user interaction history, scalability optimizations for cross-encoder inference, and aspect-based sentiment modeling for fine-grained recommendation control.